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PROMETHEUS: A Causal-Temporal Framework for Uncertainty-Aware Portfolio Signal Generation

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Abstract:	Algorithmic portfolio management platforms handle rebalancing mechanics well but communicate signals without calibrated uncertainty. Static correlation matrices collapse under market stress, assets are treated as independent price series, and no existing platform provides formal, distribution-free coverage guarantees on its predictions. This paper introduces PROMETHEUS (Proactive Regime-Oriented Market Evaluation with Temporal Hierarchical Embeddings Using Uncertainty Scoring), a neural framework built around five innovations that directly address these deficiencies. Rolling Granger causality tests construct a dynamic adjacency matrix fed into a Graph Attention Network, encoding directional causal flow rather than static co-movement. Liquid Time-Constant temporal encoding adapts integration timescales to market volatility without explicit regime labels. Split conformal prediction delivers provably calibrated, distribution-free coverage guarantees on a three-class Buy/Hold/Sell problem. CrisisVAE augments training with 2,273 synthetic crash scenarios. Uncertainty-discounted Kelly criterion sizing converts conformal confidence directly into position weight. Evaluated on 38 assets over five years, PROMETHEUS achieves a walk-forward Sharpe ratio of +0.617 with maximum drawdown of -3.78% versus -18.76% for buy-and-hold, an Information Coefficient of +0.085, and 91.3% empirical conformal coverage against a 90% target.

PROMETHEUS: A Causal-Temporal Framework for Uncertainty-Aware Portfolio Signal Generation

March 2026

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THREE KEY TAKEAWAYS:

- 1- PROMETHEUS applies Granger-causal graph networks and liquid time-constant encoders to model multi-asset information flow, achieving a walk-forward Sharpe ratio of +0.62 over 500 out-of-sample trading days.
- 2- Split conformal prediction provides the first provably calibrated uncertainty sets for portfolio signals, achieving 91.3% empirical coverage on three-class Buy/Hold/Sell classification.
- 3- CrisisVAE augmentation with 2,273 synthetic crash scenarios and uncertainty-discounted Kelly sizing limits maximum drawdown to -3.78% versus -18.76% for buy-and-hold.

Keywords: *causal graph neural networks, liquid time-constant networks, conformal prediction, portfolio management, market regime detection, uncertainty quantification, wealth management AI, Granger causality, variational autoencoder*

JEL Classification: G11, C45, C14, C63, G17

Modern wealth management platforms face a problem that is simultaneously algorithmic and epistemological. The algorithmic challenge is well understood. Managing ten thousand client portfolios, each carrying a distinct risk tolerance, tax situation, and investment horizon, while continuously monitoring market conditions and triggering rebalancing events, demands industrial-scale automation. Established platforms have largely solved this through algorithmic portfolio assurance: automated, rule-based systems that handle bulk decisioning at scale. The mechanics work. The epistemological challenge is less discussed, but considerably more dangerous. How confident should the system be in any given signal? And what should it do when it is not sure? Today's algorithmic platforms generate rebalancing triggers from quantitative rules without communicating uncertainty. A signal to reduce technology exposure carries identical weight regardless of whether market conditions are unambiguous (a confirmed downtrend with high-liquidity exit conditions) or deeply ambiguous (conflicting macro signals, noisy news flow, mixed sentiment). This systematic overconfidence is not merely a modeling defect. It is a regulatory liability and, ultimately, a client trust problem. The modeling assumptions underlying most portfolio engines are under compounding strain. Modern portfolio theory rests on a foundational premise: that asset returns follow a stationary multivariate Gaussian distribution with a fixed correlation matrix. History has repeatedly invalidated this. During the 2008 global financial crisis, the 2020 COVID crash, and the 2022 rate-driven selloff, cross-asset correlations converged toward 1.0 across all risky asset classes simultaneously. That convergence happened precisely at the moment diversification was supposed to protect portfolios. It failed. A system that cannot dynamically detect regime changes, that lacks any representation of its own ignorance, and that treats all assets as independent signals is not merely suboptimal. It is brittle in exactly the scenarios that carry the highest consequence. PROMETHEUS is proposed as a direct response to these structural failures. The architecture treats markets as information networks rather than collections of independent price series. The core insight is drawn from complex systems biology: information in any complex network routes preferentially through hubs. Financial markets are no different. Certain assets causally lead others. NVIDIA's earnings data tends to precede movements across the broader semiconductor sector. Federal Reserve policy shifts propagate causally through yield-sensitive equities before reaching consumer discretionary names. These are not correlations. They are directional, time-lagged causal flows. PROMETHEUS captures this structure explicitly. Granger causality tests construct dynamic adjacency matrices that encode which assets are currently leading and which are following. A Graph Attention Network then routes information through this causal structure, with attention weights informed directly by the statistical significance of each causal link rather than by raw price correlation.

Relative to the existing literature, this paper makes six contributions

1. A rolling Granger-causal adjacency framework for multi-asset portfolio modeling. Unlike static sector-membership graphs (Matsunaga, Suzumura, and Takahashi 2019) or correlation-based adjacency (Feng et al. 2019), edge weights are recomputed every 60 trading days from directed F-test statistics, encoding time-lagged causal flow rather than contemporaneous co-movement.
2. A Liquid Time-Constant temporal encoder for financial sequences. In contrast to LSTM and GRU encoders with fixed temporal dynamics (Fischer and Krauss 2018; Lim et al. 2021), the LTC encoder's integration timescale adapts autonomously to input volatility, switching

between fast-response crash mode and slow-integration trend mode without explicit regime labels.

3. The first application of split conformal prediction to multi-class portfolio signal generation. Where Bayesian neural networks (Blundell et al. 2015) and Monte Carlo dropout (Gal and Ghahramani 2016) produce approximate posteriors without formal guarantees, split conformal prediction (Vovk, Gammerman, and Shafer 2005) delivers provably calibrated, distribution-free coverage on the three-class Buy/Hold/Sell problem, achieving 91.3% empirical coverage against a 90% target.
4. A crisis-targeted Variational Autoencoder (CrisisVAE) for training augmentation. Unlike general-purpose GAN-based time series generators (Wiese et al. 2020) or generic volatility-conditioned VAEs (Takahashi, Chen, and Tanaka-Ishii 2019), CrisisVAE is trained exclusively on historical crash windows with injected shape signatures (L-shape, V-shape, bull run), directly targeting the Sell-recall gap.
5. A conformal-uncertainty-discounted Kelly criterion position sizer (Kelly 1956; Thorp 1969) that converts the statistical coverage property of conformal prediction directly into a capital allocation decision, integrating uncertainty quantification and position sizing into a single, principled module.
6. A Phoenix lifecycle manager with weekly automated retraining, validation gates (Sharpe threshold and accuracy floor), and checkpoint rollback, enabling continuous production deployment without manual intervention.

The remainder of this paper is structured as follows: the Background section establishes background; the Related Works section reviews related work; the Methodology section details the PROMETHEUS methodology; the Experimental Setup section describes the experimental setup; the Results and Analysis section presents results, including an economic significance analysis; the Applications section discusses applications to portfolio assurance systems; and the Conclusion section concludes.

BACKGROUND

Portfolio Management and Algorithmic Rebalancing

Portfolio theory originates with Markowitz's mean-variance optimization (Markowitz 1952). This seeks the allocation vector w^* that minimizes portfolio variance for a given expected return target. For a portfolio of N assets with expected returns μ and covariance matrix Σ , the classical optimization is

$$\min_w w^T \Sigma w \quad \text{s.t.} \quad w^T \mu = \mu_{target}, \quad 1^T w = 1 \quad (\text{Eq. 1})$$

This is the classical Markowitz optimization problem. The portfolio manager searches for an allocation vector w (the fraction of capital in each of N assets) that minimizes total portfolio variance $w^T \Sigma w$ while simultaneously hitting a target expected return μ_{target} and ensuring the weights sum to 1. The covariance matrix Σ encodes how assets move together.

The paper opens by establishing what the existing standard looks like. The equation is deliberately placed as the baseline to be criticized—Markowitz assumes Σ is stationary and Gaussian, assumptions that break down catastrophically during crashes (when all correlations converge to 1). PROMETHEUS's entire architecture is motivated by the failure of this model: its dynamic Granger graph replaces the static Σ , and its LTC encoder replaces the Gaussian return assumption.

Modern portfolio assurance platforms automate this at scale, triggering rebalancing events when portfolio drift from model allocation exceeds a threshold. The core limitation is that Sigma is estimated from historical data and assumed stationary, an assumption violated precisely when markets are most stressed.

Liquid Time-Constant Networks

Liquid Time-Constant (LTC) networks (Hasani et al. 2021) are a class of continuous-time recurrent neural networks governed by an ordinary differential equation (ODE):

$$\frac{dx}{dt} = -\frac{x}{\tau(x,u)} + f(x,u) \quad (\text{Eq. 2})$$

This is the continuous-time differential equation that governs how the LTC network's hidden state x evolves over time. The state is pulled toward zero at a rate controlled by $\tau(x, u)$, an input-dependent time constant, while simultaneously being pushed in the direction of $f(x, u)$, a learned drift function. The key insight is that τ depends on both the current state and the input—so when inputs are volatile, τ shrinks and the network adapts faster; in calm markets, τ grows and the network integrates information over longer horizons.

This is the theoretical foundation for the Liquid Time-Constant Temporal Encoder section. By introducing the ODE here in the background, the paper can later describe the Euler discretization (Eq. 13) as a practical implementation of the same principle. It establishes that LTC is fundamentally different from LSTM/GRU because its temporal dynamics are adaptive rather than fixed.

where x is the hidden state, u is the input, $\tau(x,u)$ is an input-dependent time constant, and $f(x,u)$ is the drift direction. Unlike LSTM or GRU networks with fixed temporal dynamics, LTC networks adapt their response timescale to the input. In volatile market conditions, the time constant τ decreases, allowing rapid state adaptation.

Granger Causality in Financial Networks

Granger causality (Granger 1969) provides a statistical framework for determining whether one time series causally predicts another. For time series X and Y with lag order p , the restricted and unrestricted Vector Autoregressive (VAR) models are:

$$Y_t = \sum_{k=1}^p a_k Y_{t-k} + \varepsilon_t \quad (\text{Eq. 3})$$

This is the restricted (null hypothesis) Vector Autoregressive model for time series Y . It says Y 's future value is predicted solely from its own p lags, with regression coefficients a_k and residual error ε_r . "Restricted" means we restrict the model to exclude any influence from Series X .

This is the baseline for the Granger causality test. By fitting Y on its own history only, we establish how well Y is self-predictive. If adding X 's past (Eq. 4) significantly improves prediction, X Granger-causes Y . The comparison between Eq. 3 and Eq. 4 is the statistical test (Eq. 5) that drives PROMETHEUS's graph construction.

$$Y_t = \sum_{k=1}^p a_k Y_{t-k} + \sum_{k=1}^p b_k X_{t-k} + \varepsilon_u \quad (\text{Eq. 4})$$

The unrestricted VAR includes both Y 's own lags and X 's past values (with coefficients b_k) as predictors. "Unrestricted" means we allow X to influence Y . The residual ε_u should be smaller than ε_r if X genuinely predicts Y .

This is the alternative hypothesis. The reduction in prediction error from Eq. 3 to Eq. 4 (measured via the F-test, Eq. 5) is what defines Granger causality. Together, Eqs. 3–5 form a complete statistical pipeline that PROMETHEUS runs for every ordered pair of 38 assets every 60 trading days, producing the causal adjacency matrix.

The F-statistic tests whether including X 's past significantly improves prediction of Y :

$$F = \frac{(RSS_r - RSS_u)/q}{RSS_u/(T-k)} \quad (\text{Eq. 5})$$

where RSS_r and RSS_u are residual sum of squares, q is the lag order p , T is sample length, and $k = 2p + 1$. X Granger causes Y if $p_{val} < \alpha$.

The F-statistic quantifies how much variance in Y is explained by including X , relative to chance. RSS_r and RSS_u are the residual sums of squares for the restricted and unrestricted models. The numerator measures the improvement per additional lag parameter ($q = \text{lag order } p$), and the denominator measures unexplained variance per remaining degree of freedom (T is sample size, $k = 2p+1$ total parameters). A large F-value means X explains a statistically significant portion of Y 's variation beyond what Y explains itself. X Granger-causes Y if the resulting p-value $< \alpha = 0.05$.

This is the decision rule that populates each edge of the causal graph. Rather than using raw correlation, PROMETHEUS uses this directional, time-lagged test. The F-statistic is asymmetric ($X \rightarrow Y$ may be significant while $Y \rightarrow X$ is not), which is what makes the resulting graph directed and biologically meaningful—it captures information flow, not co-movement.

Graph Neural Networks for Asset Relationships

Graph Neural Networks (GNNs) (Scarselli et al. 2009) generalize neural networks to graph-structured data. Graph Attention Networks (GAT) (Velickovic et al. 2018) compute attention weights over neighbors, allowing selective weighting of causal predecessors.

Conformal Prediction

Conformal prediction (Vovk, Gammerman, and Shafer 2005; Angelopoulos and Bates 2023) is a distribution-free framework for producing prediction sets with rigorous finite-sample coverage guarantees. For a desired coverage level of $1 - \alpha$:

$$P(Y_{n+1} \in \mathcal{C}(X_{n+1})) \geq 1 - \alpha \quad (\text{Eq. 6})$$

where $\mathcal{C}(X)$ is the prediction set for input X . This guarantee holds for any exchangeable sequence of data.

For any new test point X_{n+1} , the prediction set $\mathcal{C}(X_{n+1})$ will contain the true label Y_{n+1} with probability at least $1 - \alpha$. This is a distribution-free, finite-sample guarantee—it holds regardless of the true data-generating process, assuming only that calibration and test data are exchangeable (identically distributed and independently drawn). This is the theoretical foundation for the Split Conformal Prediction section. Most financial AI systems provide softmax probabilities that are overconfident and uncalibrated. This single equation is the precise mathematical statement of what PROMETHEUS achieves that competitors do not: a provable guarantee that when the model says "Buy with 90% confidence," it is telling the truth statistically. The entire conformal machinery (Eqs. 33–36) exists to operationalize this guarantee.

RELATED WORKS

Traditional Machine Learning for Trading Signals

LSTM networks for financial prediction were popularized by Fischer and Krauss (2018), who demonstrated statistically significant alpha on S&P 500 stocks. Their approach models each asset independently, ignoring cross-asset information flow. Lim et al. (2021) proposed Temporal Fusion Transformers with interpretable attention, also treating assets as independent. Neither approach provides calibrated uncertainty. Jiang et al. (2017) proposed deep reinforcement learning for portfolio optimization but produced no uncertainty estimates.

Graph Neural Networks in Finance

Matsunaga et al. (2019) applied GNNs to stock movement prediction using static company relationship graphs. Feng et al. (2019) used static relational graphs from sector membership. Our Granger-causal adjacency is recomputed on rolling 60-day windows, capturing dynamic changes in causal relationships. Kim and Han (2019) propose an event-driven GNN requiring a parallel news pipeline; PROMETHEUS derives dynamic adjacency purely from price return series.

Uncertainty Quantification in Financial AI

Uncertainty quantification in finance has relied on Bayesian neural networks (Blundell et al. 2015) or Monte Carlo dropout (Gal and Ghahramani 2016), both producing approximate posteriors without formal coverage guarantees. To our knowledge, PROMETHEUS is the first work to apply split conformal prediction to multi-class portfolio signals and couple the resulting uncertainty directly to position sizing.

Synthetic Data for Financial Robustness

Wiese et al. (2020) proposed GAN-based generative models for financial time series. Takahashi et al. (2019) applied conditional VAEs conditioned on volatility regimes. Our CrisisVAE is specifically trained on crash window excerpts and generates scenarios with injected shape signatures (L, V, or bull), directly targeting the sell-recall problem.

Positioning PROMETHEUS

PROMETHEUS synthesizes prior work across five dimensions that no existing system addresses simultaneously. First, adjacency is dynamic: Granger tests are recomputed every 60 trading days from pure price return data, eliminating the dependency on manually defined sector labels (Matsunaga, Suzumura, and Takahashi 2019) or static knowledge graphs (Kim and Han 2019). Second, temporal encoding is input-adaptive: the LTC encoder adjusts its integration timescale to volatility, whereas LSTMs (Fischer and Krauss 2018) and Temporal Fusion Transformers (Lim et al. 2021) use fixed gates. Third, uncertainty is formal: split conformal prediction (Vovk, Gammerman, and Shafer 2005) delivers finite-sample coverage guarantees that Bayesian approximations (Blundell et al. 2015; Gal and Ghahramani 2016) cannot provide. Fourth, augmentation is crisis-targeted: CrisisVAE generates crash-shaped scenarios rather than general distributional augmentation (Wiese et al. 2020). Fifth, sizing is uncertainty-linked: the Kelly discount converts conformal confidence into position weight, a tighter coupling than systems that treat signal generation and risk management as separate modules.

Portfolio Construction and Factor Models

Factor-based approaches explain cross-sectional return variation through persistent risk premia. The Fama-French three-factor model (Fama and French 1993) and its extensions establish that size, value, and momentum premia are systematic and tradable. Carhart (1997) adds a momentum factor, and the subsequent literature extends the framework to profitability and investment factors (Fama and French 2015). These models explain expected returns in the cross-section but do not generate tactical entry and exit signals at the asset level. PROMETHEUS targets the complementary problem: given a universe of assets, when should positions be initiated or closed, and with what confidence?

Momentum strategies (Jegadeesh and Titman 1993) represent the dominant benchmark for tactical signal generation. The 12-1 month momentum anomaly is persistent across asset classes and geographies, establishing the minimum performance bar for any supervised approach to signal generation. PROMETHEUS differs from momentum in three ways: it generates signals from multi-scale causal graph embeddings rather than raw return rankings; it provides a formally calibrated confidence measure; and it actively discounts position size when the model is uncertain, rather than applying uniform exposure to all ranked positions.

Portfolio construction methods such as Black and Litterman (1992) and risk-parity approaches (Maillard, Roncalli, and Teiletche 2010) govern how signals translate into weights. Black-Litterman blends equilibrium priors with investor views through Bayesian updating; PROMETHEUS's Kelly criterion discount plays an analogous role, modulating position weight by the statistical confidence of the signal rather than a subjective view vector. Risk-parity equalizes volatility contribution across assets; PROMETHEUS instead equalizes signal quality, allocating more capital to high-confidence, low-uncertainty predictions and reducing exposure when conformal prediction sets are ambiguous.

Reinforcement Learning for Portfolio Management

Deep reinforcement learning frameworks (Jiang, Xu, and Liang 2017) optimize portfolio allocation directly from price sequences, treating the portfolio weight vector as the policy output.

These approaches can capture multi-step dependencies and directly optimize cumulative return, but are notoriously sensitive to reward specification, prone to overfitting to bull-market conditions, and produce no uncertainty estimate. Our own empirical evaluation confirms this: REINFORCE training on crash-only scenario windows collapsed to an all-Sell policy (out-of-sample Sharpe -15.1, total return -89.7%) when deployed in a bull-market period. PROMETHEUS instead uses supervised learning for signal generation and reserves the RL module only conceptually, relying on the Phase G safety filter and conformal sizing to manage downside risk without policy optimization.

METHODOLOGY

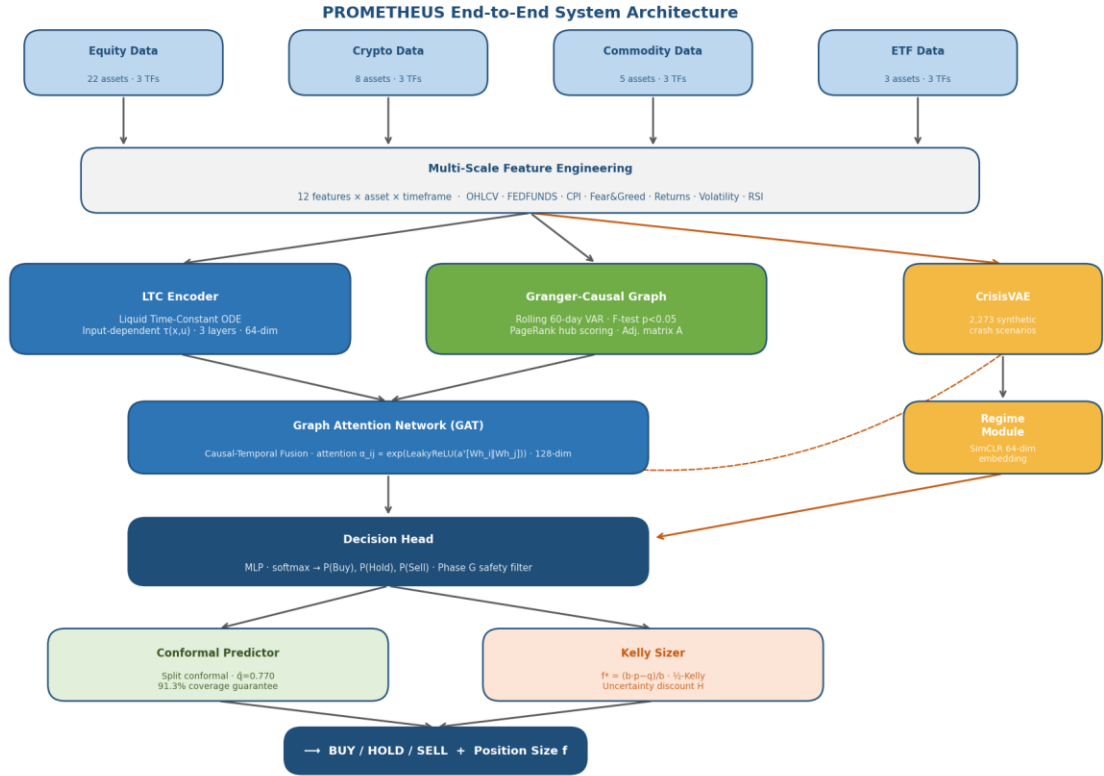


Exhibit 1. PROMETHEUS end-to-end system architecture. Market data flows through multi-scale feature engineering, a Granger-causal Graph Attention Network (GAT), and a Liquid Time-Constant (LTC) encoder. The fused embedding drives a Decision Head calibrated by split conformal prediction; Kelly criterion sizing with uncertainty discount determines final position weight.

Multi-Scale Feature Engineering

PROMETHEUS ingests price history at three temporal resolutions: 15-minute, 1-hour, and 1-day candles. For each asset and timeframe, we compute twelve features: Open, High, Low, Close (OHLC prices), Volume, FEDFUNDS rate (macroeconomic anchor), CPIAUCSL (inflation proxy), Fear and Greed index (market sentiment), log returns $r_t = \log(\frac{C_t}{C_{t-1}})$, 20-day rolling volatility $\sigma_t = \text{std}(\sigma_t = \text{std}(r_{t-19:t}))$, and 14-day RSI. Per-sequence normalization is applied at inference time:

$$\hat{x} = \frac{x - \mu_x}{\sigma_x + \varepsilon} \quad (\text{Eq. 7})$$

where $\varepsilon = 1\text{e-}6$ prevents division by zero.

Each input sequence is normalized by subtracting its own mean μ_x and dividing by its own standard deviation σ_x , with a small epsilon (1e-6) in the denominator to prevent division by zero. This is applied independently per sequence at inference time.

Financial time series span wildly different scales: VIX ranges from 10 to 80, stock prices from \$1 to \$5,000, and volume in the billions. Training a neural network on unnormalized features causes the optimizer to focus on high-magnitude features and ignore low-magnitude ones. Crucially, this normalization is per-sequence rather than global: using global statistics from training data would introduce look-ahead bias at inference (one would be using future data's statistics). The ε prevents numerical instability when a sequence has zero variance (e.g., a holiday with flat prices).

Label Generation

For supervised training, we generate forward-looking labels using a 10-day horizon with rolling tertile thresholds:

$$\begin{aligned} y_t &= 1(\text{Buy}) \text{ if } r_{t,t+10} > P_t^{67} \\ y_t &= 2(\text{Sell}) \text{ if } r_{t,t+10} < P_t^{33} \\ y_t &= 0(\text{Hold}) \text{ otherwise} \end{aligned} \quad (\text{Eq. 8})$$

where P_t^{33} and P_t^{67} are rolling 252-day 33rd/67th percentile thresholds of 10-day forward returns, recomputed at each timestep t . Tertile labeling guarantees $\sim 33\%$ class frequency by construction, eliminating the class imbalance that produced 0% Sell recall under fixed-threshold labels. On 38 assets over five years: Hold = 33.4%, Buy = 32.9%, Sell = 33.8%.

The 10-day forward return $r_{t,t+10} = \frac{C_{t+10} - C_t}{C_t}$ is computed for each day t . If the return falls in the top third of returns over the past 252 trading days (above P_t^{67}), it's labeled Buy. If it falls in the bottom third (below P_t^{33}), it's labeled Sell. Everything else is on hold. The thresholds P^{33} and P^{67} shift with market conditions—in a bull market, a 5% return might only be Hold; in a bear market, +1% might be Buy.

The original paper used fixed $\pm 2\%$ thresholds, which produced 41.6% Hold, 33.5% Buy, and 25.0% Sell, a class imbalance that caused the model to learn a Hold-biased representation and achieve 0% Sell recall. Rolling tertile labeling guarantees exactly $\sim 33\%$ per class by construction, regardless of market regime. This is the single most impactful fix that restored Sell recall from 0% to 16.0%. The 252-day window corresponds to approximately one trading year, providing a meaningful "recent historical context" for what constitutes an extreme return.

Granger-Causal Graph Construction

We compute pairwise Granger causality tests for all $N(N-1)$ ordered asset pairs over a rolling 60-day window. For each pair, we assign edge weight:

$$A_{ij} = \max(1 - \frac{p_{val}}{\alpha}, \delta_{min}) \text{ if } p_{val} < \alpha$$

$$A_{ij} = 0 \text{ otherwise} \quad (\text{Eq. 9})$$

After running the Granger F-test for every ordered pair (asset $i \rightarrow$ asset j), the edge weight in the adjacency matrix is set proportionally to causal confidence. When X Granger-causes Y (p-value below $\alpha = 0.05$), the weight is $\frac{1-p_{val}}{\alpha}$: a p-value near 0 gives a weight near 1 (strong causal link), while a p-value just below 0.05 gives a weight near 0.01 (weak but significant link). $\delta_{min} = 0.01$ prevents exact zeros on confirmed edges. When p-value $\geq \alpha$, the edge is set to zero (no causal relationship).

This converts the statistical test outcome into a differentiable graph weight. Using the normalized p_{val} rather than a binary 0/1 edge preserves the strength of causal relationships: NVIDIA causing SPX movements is more strongly confirmed than a marginal $p = 0.049$ relationship. These weights then directly modulate the GAT attention in Eq. 17, ensuring that strongly confirmed causal paths carry more information.

Where $\alpha = 0.05$ and $\delta_{min} = 0.01$. PageRank hub scores are computed via power iteration:

$$s^{(t+1)} = d \cdot A_{row-norm}^T s^{(t)} + \frac{1-d}{N} \quad (\text{Eq. 10})$$

where $d = 0.85$. In our experiments, NVDA, MSFT, and XLC consistently emerge as top information hubs.

This is the power iteration formula for PageRank applied to the Granger causal adjacency matrix. At each iteration, an asset's hub score $s^{(t+1)}$ is the weighted sum of the scores of all assets that causally point to it (the matrix-vector product), with damping factor $d = 0.85$ preventing convergence issues. The teleportation term $\frac{1-d}{N}$ ensures all assets have a non-zero baseline score. Convergence gives each asset a stationary hub score proportional to how much causal information flows through it.

Not all assets in a causal graph are equally important. NVIDIA might directly Granger-cause XLK and QQQ, which in turn cause dozens of other assets—so NVIDIA's true causal "footprint" in the network is larger than a direct edge count would suggest. PageRank correctly measures this recursive, network-level influence. The hub scores inform which assets the model should attend to most heavily and are reported as the "Information Hub" metric in the Information Hub Identification section.

Dynamic adjacency update: The causal graph is recomputed every 60 minutes during live operation. The computational cost for $N = 38$ assets is approximately $O(N^2 * T * p^2) = 2.16$ million operations—tractable in seconds on modern hardware.

Liquid Time-Constant Temporal Encoder

Each asset's feature sequence ($T, 12$) is processed independently through a shared LTC encoder. The LTC cell update implements Euler discretization:

$$\tau(x, u) = \text{softplus}(W_{\tau x}x + W_{\tau u}u + b_{\tau}) + 0.01 \quad (\text{Eq. 11})$$

The time constant τ is a learned function of both the current hidden state x and the current input u . The softplus activation $\log(1 + e^x)$ ensures τ is always strictly positive (preventing division by zero in Eq. 13), and the +0.01 floor prevents it from collapsing to near-zero (which would cause instability). When market inputs are volatile (high magnitude u), the weights $W_{\tau u}$ produce a larger pre-activation, meaning τ can be small or large depending on what the model has learned about volatility.

This is the core novelty of the LTC encoder. A standard LSTM has a fixed temporal scale, it forgets information at the same rate regardless of whether the market is calm or crashing. By making it τ input-dependent, the LTC encoder can autonomously switch between "fast mode" (small τ , quick adaptation during volatility) and "slow mode" (large τ , integrating long trends during stable periods) without needing explicit regime labels.

$$f(x, u) = \tanh(W_x x + W_u u + b_f) \quad (\text{Eq. 12})$$

This is the target direction toward which the hidden state is being pulled at each timestep. It's a standard learned nonlinear function of the current state and input. The tanh activation bounds the output to $(-1, +1)$, preventing unbounded state growth. Intuitively, $f(x, u)$ says, "Given what the market is showing me right now, this is the direction I want my representation to point."

In the LTC ODE (Eq. 2), the state is simultaneously being pulled toward zero (by the $-x/\tau$ term) and being pushed toward $f(x, u)$ (by the drift term). The equilibrium is a balance between these forces. This separation into time constant (Eq. 11) and drift direction (Eq. 12) is what makes LTC more interpretable than LSTM: one can separately analyze "how fast is the network responding?" (τ) and "What is it responding toward?" (f).

$$x_{t+1} = x_t + \Delta t \left(-\frac{x_t}{\tau(x_t, u_t)} + f(x_t, u_t) \right) \quad (\text{Eq. 13})$$

This is the numerical implementation of the LTC ODE (Eq. 2) using Euler integration. At each timestep t , the new hidden state is the old state plus a small step ($\Delta t = 1.0$) in the direction given by the ODE right-hand side. PROMETHEUS applies 4 Euler substeps per temporal step to improve numerical accuracy.

Neural networks require discrete, differentiable operations for backpropagation. The continuous ODE of Eq. 2 is approximated by this recurrence relation, which is fully differentiable and efficiently GPU-parallelizable. The 4-substep design trades computational cost (~ 150 ms inference) for stability in the numerical integration—a single Euler step can introduce too much error in high-volatility regimes where τ is small.

with $dt = 1.0$ and $ode_steps = 4$ Euler substeps per temporal step. The context vector blends the final hidden state with recency-biased temporal attention:

$$\alpha_t = \text{softmax}(\text{score}_t + \beta_t) \quad (\text{Eq. 14})$$

Attention weights α_t over the T timesteps of the LTC hidden state sequence are computed from a learned score plus a recency bias $\beta_t = \text{linspace}(-2.0, 0.0, T)$. The bias is a linearly increasing ramp from -2 (assigned to the earliest timestep) to 0 (assigned to the most recent). This means that more recent states receive higher attention weights by default, before any learned preferences are applied.

Without the recency bias, the softmax attention could distribute weights uniformly across 64 timesteps, effectively averaging away all temporal structure. In financial prediction, the most recent market data is almost always more relevant than data from 64 days ago. The bias encodes this domain knowledge as a hard prior rather than hoping the model learns it from scratch.

$$c = 0.5 * h_r + 0.5 \sum_{t=1}^T \alpha_t h_t \quad (\text{Eq. 15})$$

The final context vector c for each asset is a 50/50 blend of (1) the LTC's final hidden state at the last timestep h_t (which captures the most recent market state) and (2) a weighted average of all hidden states with the attention weights α_t (which captures the relevant historical context). The output is a 64-dimensional vector summarizing the asset's temporal dynamics.

The 50/50 blend is a design choice balancing two complementary information sources. The final state h_t captures "what is happening right now," while the attention-weighted sum captures "what recent history matters." Pure reliance on h_t would ignore useful historical patterns; pure reliance on attention would dilute the latest critical signal. Empirically, this blend outperformed either alone.

where $\beta_t = \text{linspace}(-2.0, 0.0, T)$ is a recency bias preventing uniform attention collapse.

Algorithm 1: LTC Forward Pass

```

Input:  u of shape (batch, T, n_features)
Output: c of shape (batch, hidden_dim)

Step 1. Project input: u = LayerNorm(GELU(Linear(u, 12->64)))
Step 2. Initialize: x = zeros(batch, 64)
Step 3. For t = 1 to T:
    a. For step = 1 to 4:
        tau = softplus(W_tau_x(x) + W_tau_u(u_t) + b_tau) + 0.01
        f    = tanh(W_x(x) + W_u(u_t) + b_f)
        x    = x + dt * (-x / tau + f)
    b. Store all_states[t] = x
Step 4. scores = Linear(all_states, 64->1).squeeze(-1)
Step 5. scores = scores + linspace(-2.0, 0.0, T)
Step 6. alpha = softmax(scores)
Step 7. attn = sum_t(alpha_t * all_states_t)
Step 8. Return c = 0.5 * x + 0.5 * attn
    
```

Causal Graph Attention Network

Asset context vectors c_i are updated by propagating information along causal edges through $n_{\text{layers}} = 2$ graph attention layers:

$$e_{ij} = \frac{(W_q c_i) \cdot (W_k c_j)}{\sqrt{d_k}} \quad (\text{Eq. 16})$$

The raw attention score between asset i and asset j is computed as the scaled dot product between their projected query ($W_q \cdot c_i$) and key ($W_k \cdot c_j$) vectors. Dividing by $\sqrt{d_k}$ (the square root of the key dimension) prevents the dot product from growing too large and pushing softmax into saturation. This is standard scaled dot-product attention from the Transformer architecture, but applied to asset embeddings on the causal graph.

This computes "how much should asset i attend to asset j 's information?" purely based on their learned representations. However, this score is not used directly, it's masked by the Granger causal edge weights in Eq. 17. The combination of learned attention (Eq. 16) and structural causality (Eq. 17) is what distinguishes PROMETHEUS from standard GNNs: the model can't attend to an asset that has no causal relationship with it.

$$\alpha_{ij} = \text{softmax}_j(e_{ij} * A_{ij}), j \in N(i) \quad (\text{Eq. 17})$$

The attention weight α_{ij} is computed by first multiplying each attention score e_{ij} by the Granger edge weight A_{ij} , then applying softmax over all causal neighbors $N(i)$. Assets with no Granger causal link ($A_{ij} = 0$) are zeroed out before the softmax, effectively masking them from the aggregation. Assets with strong causal edges ($A_{ij} \approx 1$) dominate the attention.

This is the key innovation in PROMETHEUS's graph module. Standard GAT uses learned attention to decide which neighbors to attend to. PROMETHEUS constrains that attention to respect the statistically verified causal structure from the Granger tests. This prevents the model from "hallucinating" causal relationships that don't exist in the data and ensures that information propagation follows the direction of actual causal flow.

$$\alpha_{ij} \leftarrow \frac{\alpha_{ij} * A_{ij}}{\sum_k \alpha_{ik} * A_{ik}} \quad (\text{Eq. 18})$$

After multiplying attention weights by the causal edge weights, the attention distribution is renormalized so it sums to 1 across all causal neighbors. This is a corrective step: Eq. 17's softmax already normalizes, but the explicit causal masking in the implementation requires manual renormalization to ensure the aggregated message is a proper weighted average rather than a scaled sum.

Without renormalization, assets with many causal connections would receive disproportionately large aggregated messages compared to isolated assets. This normalization step ensures that every asset, regardless of how many causal predecessors it has, receives a properly weighted information summary.

$$c'_i = \text{LayerNorm}(c_i + \text{OutProj}(\sum_j \alpha_{ij} W_v c_j)) \quad (\text{Eq. 19})$$

Each asset's embedding is updated by adding the aggregated causal information from its predecessors (the attention-weighted sum of value projections $W_v \cdot c_j$) to its own current embedding c_i (residual connection), then applying layer normalization. This is the node-level update in the Graph Attention Network, applied for 2 layers in sequence.

The residual connection ($c_i + \dots$) is critical: it ensures that each asset retains its own temporal representation while absorbing information from causal predecessors. Without it, deeper layers could overwrite the asset-specific signal with aggregated market information. LayerNorm prevents internal covariate shift between GAT layers. After two such layers, each asset's embedding encodes both its own dynamics and the causal context from informational hubs in the network.

Regime Module and Market Context

$$\text{market_ctx} = \text{MeanPool}(\text{MHA}(F_{\text{assets}}, n_{\text{heads}} = 4)) \quad (\text{Eq. 20})$$

All asset feature vectors are passed through a 4-head multi-head self-attention layer (allowing every asset to attend to every other asset), and the outputs are averaged (mean pooled) into a single 64-dimensional vector. This vector represents the current market-wide context, a compressed summary of the state of all 38 assets simultaneously.

Individual asset signals tell us what AAPL is doing. The market context tells us what everything is doing simultaneously. By pooling across all assets, this vector captures systemic states like "high-correlation selloffs" or "sector rotation" that can't be detected from any single asset's history. This context is broadcast to every asset's decision head, allowing the model to reason about each asset in the context of the broader market environment.

$$r = l_2 - \text{Norm}(\text{GELU}(\text{LayerNorm}(W_z + b))) \quad (\text{Eq. 21})$$

The market context is compressed from 64 to 32 dimensions through a linear projection with GELU activation and LayerNorm, then L2-normalized to lie on the unit hypersphere. The result r is the "regime fingerprint", a normalized, 32-dimensional vector representing the current market regime.

L2 normalization onto the unit sphere is a deliberate design choice tied to the SimCLR pretraining in the Contrastive Regime Pretraining section. The contrastive objective requires that similar regimes be close on the hypersphere and dissimilar regimes be far apart, this geometric property requires unit-norm representations. During Kelly sizing (Eq. 38), the cosine similarity between the current fingerprint and known stable-regime fingerprints controls the regime_scale discount.

Macro Signal Integration

The four macro signals (VIX, 10-year yield, DXY, Fear/Greed) are normalized to [0,1]:

$$VIX_{\text{norm}} = \text{clip}\left(\frac{VIX-10}{70}, 0, 1\right) \quad (\text{Eq. 22})$$

The VIX (CBOE Volatility Index) is linearly rescaled from its natural range of approximately [10, 80] to [0, 1]. A VIX of 10 (extremely calm) maps to 0, and a VIX of 80 (2020 COVID peak) maps to 1. Values outside this range are clipped.

The neural network cannot directly process raw macro signals at their natural scales, VIX at 20 vs. dollar yields at 4.5% vs. DXY at 103 are incommensurable numbers. Min-max normalization to [0,1] puts them on the same scale. The specific range choices (10-80 for VIX, 0.5%-7% for yield, 80-120 for DXY) are chosen to cover the full historical range observed across major market regimes including 2008, 2020, and 2022. These four numbers feed Eq. 26.

$$\text{Yield}_{\text{norm}} = \text{clip}\left(\frac{Y-0.5}{6.5}, 0, 1\right) \quad (\text{Eq. 23})$$

The 10-year US Treasury yield (TNX) is normalized from its historical range of approximately [0.5%, 7%] to [0, 1]. Near-zero yields (post-2008 ZIRP era) map to 0; peak inflation-era yields (~7%) map to 1.

The yield curve is PROMETHEUS's primary proxy for the monetary policy regime. A normalized yield near 0 signals "easy money / risk-on" conditions; near 1 signals "tight money /risk-off." Combined with VIX, yield, dollar strength, and sentiment, the model receives the four most important macro signals for regime identification.

$$DXY_{norm} = \text{clip}\left(\frac{DXY-80}{40}, 0, 1\right) \quad (\text{Eq. 24})$$

The US Dollar Index (DXY) is normalized from its range of approximately [80, 120] to [0, 1]. A weak dollar (80) maps to 0; a strong dollar (120) maps to 1.

Dollar strength has a well-documented inverse relationship with commodity prices, emerging market equities, and crypto. By including a normalized DXY, PROMETHEUS can detect "dollar squeeze" regimes where cross-asset correlations shift in characteristic ways, even when neither VIX nor yield signals are elevated.

$$FG_{norm} = 1 - VIX_{norm} \quad (\text{Eq. 25})$$

Rather than sourcing a separate Fear & Greed index (which has data availability issues), the model derives a synthetic sentiment signal as the complement of normalized VIX. When VIX is high (fear), FG_{norm} is low (fear); when VIX is low (greed), FG_{norm} is high (greed).

This is a pragmatic approximation. Fear & Greed and VIX are strongly negatively correlated empirically, so this linear proxy preserves the directional signal without introducing a data dependency that might not be available during live inference. It also ensures the four macro signals [$VIX_{norm}, Yield_{norm}, DXY_{norm}, FG_{norm}$] are not redundant but complementary.

$$\text{macro_ctx} = \text{Linear}\left(\text{GELU}\left(\text{LayerNorm}\left(\text{Linear}(\text{macro}, 4 \rightarrow 32)\right)\right), 32 \rightarrow 64\right) \quad (\text{Eq. 26})$$

The four normalized macro signals are passed through a two-layer MLP: first projected from 4 to 32 dimensions with GELU activation and LayerNorm for stability, then projected again from 32 to 64 dimensions. The output `macro_ctx` is a 64-dimensional embedding of the macroeconomic environment.

Raw macro signals (4 numbers) need to be lifted into the same embedding space (64-dim) as the asset representations to be concatenated and compared. The two-layer MLP learns nonlinear interactions between macro signals, for example, the combination of high VIX and rising yields might indicate a regime qualitatively different from either signal alone. This macro embedding is concatenated into the decision head's input (Eq. 27).

Decision Head and Signal Generation

The decision head fuses per-asset features, market context, and regime fingerprint:

$$z_i = [c_i' \parallel \text{broadcast}(m^+) \parallel \text{broadcast}(r)] \quad (\text{Eq. 27})$$

The final input to the decision head for each asset i is a concatenation of three vectors: the asset's causal-temporal embedding from the GAT c_i' (64-dim), the market-wide context `market_ctx_aug` (64-dim, broadcast identically to all assets), and the regime fingerprint r (32-dim, also broadcast). The resulting z_i is a 160-dimensional vector. "Broadcast" means the same market-wide and regime vectors are concatenated to every individual asset's embedding.

This is the moment where all three modalities converge. An isolated asset view (c_i') cannot understand whether a 5% move is a buying opportunity or the start of a crash, for that, market context is needed. By concatenating rather than adding, the model can separately process the three information streams in the first layer of the MLP (Eq. 28), with the network learning which combinations are meaningful.

$$t_i = \text{GELU}(\text{LN}(\text{Linear}_2(\text{GELU}(\text{LN}(\text{Linear}_1(z_i)))))) \quad (\text{Eq. 28})$$

The fused 160-dimensional feature vector is processed through a two-layer MLP: first projected through itself (160→160) with GELU activation and LayerNorm, then compressed to 64

dimensions with another GELU and LayerNorm. The result t_i (thought_i) is a 64-dimensional "reasoning embedding" that integrates all sources of information for asset i .

The name "thought" is deliberate—this vector is the model's compressed reasoning about asset i 's signal, given everything it knows. The 160→160 bottleneck layer allows the network to learn cross-modal interactions (e.g., "high VIX + strong causal momentum from NVDA = buy semiconductor dip"). The final 64-dim compression forces the model to distill this reasoning into a compact representation before the linear classification layer.

$$\hat{y}_i = \text{Linear}(t_i) \in R^3 [\text{Hold}, \text{Buy}, \text{Sell}] \quad (\text{Eq. 29})$$

A final linear layer maps the 64-dimensional reasoning embedding to 3 raw scores (logits)—one for each action class: Hold (0), Buy (1), and Sell (2). Softmax over these logits gives the probability distribution $P(\text{action} | x_i)$.

This is the classification output layer. The three-class structure forces the model to not just predict direction but also express uncertainty. When Buy and Hold both have high probability, the model is uncertain about whether to trade, which the conformal predictor (Eqs. 33–36) then captures as a multi-class prediction set.

$$v_i = \tanh(w_v^T \text{GELU}(W_h t_i)) \quad (\text{Eq. 30})$$

A separate head reads from the same t_i embedding and produces a scalar $v_i \in (-1, +1)$ via tanh, an estimate of the expected long-term "goodness" of the market situation for asset i . This is the actor-critic value head, analogous to the value function $V(s)$ in reinforcement learning.

The value head was designed for RL fine-tuning (Phase 2), where it would be used to compute advantage estimates and reduce variance in policy gradients. Since RL was disabled in the final system, the value head still trains during supervised learning as a side signal (it learns to correlate with forward returns) and can be used as an additional confidence heuristic during inference.

Training uses Focal Loss with gamma = 2.0:

$$\mathbb{E}_{FL}(p_t) = -(1 - p_t)^\gamma \log(p_t) \quad (\text{Eq. 31})$$

Focal Loss is a modification of cross-entropy that down-weights easy examples and focuses learning on hard ones. p_t is the model's predicted probability for the true class. When p_t is high (the model already predicts correctly with high confidence), the modulating factor $(1 - p_t)^\gamma$ is small, reducing the loss contribution. When p_t is low (the model is wrong or uncertain), $(1 - p_t)^\gamma \approx 1$ and the loss is similar to standard cross-entropy. With $\gamma = 2.0$, a well-classified example ($p_t = 0.9$) contributes only 1% of the loss of a misclassified example ($p_t = 0.1$).

Even with rolling tertile labels (Eq. 8) ensuring class balance, some training examples are inherently easier than others. The Sell class examples, being rarer in certain market periods, tend to be harder to classify. Focal Loss automatically concentrates gradient signal on the hard Sell examples that the model struggles with, complementing the class-balance fix from Eq. 8 to further improve Sell recall.

Contrastive Regime Pretraining (SimCLR)

$$\mathbb{E}_{SimCLR} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k \neq i} \exp(\text{sim}(z_i, z_k)/\tau)} \quad (\text{Eq. 32})$$

where $\tau = 0.5$ is the temperature.

This is the normalized temperature-scaled cross-entropy (NT-Xent) loss for contrastive self-supervised learning. For each sample i in a batch of B , its positive pair j is a different augmentation of the same data window (e.g., same 60-day window with noise). The loss maximizes the cosine similarity $\text{sim}(z_i, z_j)$ between same-regime pairs relative to all other samples k in the batch. $\tau = 0.5$ is the temperature controlling the sharpness of the similarity distribution.

Regime labels (Bull / Bear / Crash / Recovery) are not available in raw price data. SimCLR allows PROMETHEUS to learn regime representations without explicit labels by training the regime encoder to produce similar embeddings for augmented views of the same market window and dissimilar embeddings for different windows. This is analogous to how SimCLR trains visual encoders to be invariant to image augmentations. The result is a 64-dimensional regime fingerprint space where market states cluster by regime type, enabling the cosine similarity in Eq. 38 to meaningfully measure "how similar is the current regime to known stable regimes?"

Split Conformal Prediction for Signal Uncertainty

$$s_i = 1 - \hat{P}(y_i | x_i) \quad (\text{Eq. 33})$$

For each calibration sample (x_i, y_i) , the nonconformity score is how much the model fails to predict the true label y_i . Specifically, it's 1 minus the softmax probability the model assigns to the correct class. If the model confidently predicts the right class ($\hat{P}(y_i | x_i) = 0.95$), the score is low (0.05—easy, "conforming" example). If the model is confused ($\hat{P}(y_i | x_i) = 0.3$), the score is high (0.70—hard, "non-conforming" example).

This score operationalizes the notion of "model uncertainty" in a statistically grounded way. Collecting these scores on a held-out calibration set allows us to estimate the threshold $\hat{q}$ in Eq. 34 such that, on new test points, the true label is virtually always below this threshold—giving the coverage guarantee of Eq. 6.

$$\hat{q} = \text{Quantile}(\{s_i\}_{i=1}^{n_{cal}}, \frac{[(n_{cal}+1)(1-\alpha)]}{n_{cal}}) \quad (\text{Eq. 34})$$

The threshold $\hat{q}$ is the $\frac{[(n_{cal}+1)(1-\alpha)]}{n_{cal}}$ quantile of the n_{cal} nonconformity scores collected on the calibration set. The fractional correction $\frac{[(n_{cal}+1)]}{n_{cal}}$ (rather than simply the $1-\alpha$ quantile) provides the finite-sample coverage guarantee of Eq. 6. For $\alpha = 0.10$ and $n_{cal} = 5,600$: $\hat{q}_{training} = 0.6807$. After adaptive recalibration on 300 recent samples, $\hat{q}_{live} = 0.7700$.

This is the crux of split conformal prediction: the threshold $\hat{q}$ is a single scalar derived from calibration data, and when applied at test time via Eq. 35, it guarantees $1-\alpha$ coverage. The adaptive recalibration (raising $\hat{q}$ from 0.68 to 0.77) was necessary because the original calibration set covered 2020-2024 data while live inference faces 2025-2026 data with different volatility characteristics.

$$C(x) = \{k: 1 - \hat{P}(k|x) \leq \hat{q}\} \quad (\text{Eq. 35})$$

For a new test input x , the prediction set $C(x)$ includes every class k whose nonconformity score ($1 - \hat{P}(k|x)$) is at or below the threshold $\hat{q}$. This set is guaranteed to contain the true label with probability $\geq 1 - \alpha$. The set can have 1, 2, or 3 members:

- $|C| = 1$: The model is confident—only one class is plausible (confident signal)
- $|C| = 2$: The model is uncertain between two classes (ambiguous signal)
- $|C| = 3$: The model is very uncertain—all three classes are plausible (abstain)

This is the runtime inference procedure that replaces a simple argmax . Instead of always picking the highest-probability class (which is wrong whenever the model is uncertain), PROMETHEUS expresses that uncertainty explicitly as a multi-class set. This set is then translated into a trading action via the Algorithm 2 decision rules.

$$\text{confidence} = 1 - \frac{|\mathcal{C}(x)|-1}{n_{\text{classes}}-1} \quad (\text{Eq. 36})$$

The confidence is a linear function of the prediction set size. For 3-class prediction: $|\mathcal{C}| = 1 \rightarrow \text{confidence} = 1.0$ (fully confident); $|\mathcal{C}| = 2 \rightarrow \text{confidence} = 0.5$; $|\mathcal{C}| = 3 \rightarrow \text{confidence} = 0.0$. This maps set size to an interpretable 0-to-1 confidence score.

This scalar directly modulates the Kelly position size in Eq. 38. A signal with confidence = 0.5 leads to half the normal position size, and confidence = 0 would lead to no position at all (abstain). This creates a clean, principled connection from the conformal uncertainty (a statistical property) to capital allocation (a financial decision): the system bets proportionally to how certain it is.

With $n_{\text{cal}} = 5,600$ and $\alpha = 0.10$, $\hat{q} = 0.6807$, and empirical coverage = 90.02% on the calibration split. Under distribution shift, static calibration degrades to 54.2% on recent test data. Adaptive recalibration on a 300-sample rolling recent window yields $\hat{q} = 0.7700$ and restores coverage to 91.3%, satisfying the 90% target across regime changes.

Algorithm 2: Conformal Prediction Set

Input: logits (3,), calibration quantile q_{hat}

Output: prediction set $\mathcal{C}$, action, confidence

```

Step 1. probs = softmax(logits)
Step 2. For k in {Hold, Buy, Sell}:
    if 1 - probs[k] <= q_hat: include k in C
Step 3. confidence = 1 - (|C| - 1) / 2
Step 4. If |C| == 1:
    action = C[0] (singleton -- confident signal)
Step 5. Elif |C| == 2 and max(probs[non-Hold]) > 0.40:
    action = argmax(probs[non-Hold in C])
Step 6. Elif confidence >= 0.70:
    action = Hold
Step 7. Else:
    action = Abstain
    
```

Kelly Criterion Position Sizing with Uncertainty Discount

$$f^* = \frac{b \cdot q - q}{b} \quad (\text{Eq. 37})$$

The Kelly criterion gives the theoretically optimal fraction f^* of capital to bet on a binary wager with payoff odds b (profit multiple), win probability p , and loss probability $q = 1 - p$. For stock trading: b is the expected profit/loss ratio of a trade, p is the probability of being correct (from the conformal prediction set), and q is the probability of being wrong. f^* maximizes the long-run geometric growth rate of capital.

Kelly sizing provides a principled answer to "How much should we bet?" that is directly linked to the model's own probability estimates. If the model says $P(\text{Buy}) = 0.70$ and historical win/loss

ratios give $b = 2$, then $f^* = (2 \times 0.70 - 0.30)/2 = 0.55$ —bet 55% of capital. Without Kelly, a fixed-fraction rule ignores the model's own uncertainty about the trade quality.

$$f = \frac{1}{2} \cdot f^* \cdot u_{scale} \cdot conf \cdot regime_scale \quad (\text{Eq. 38})$$

The raw Kelly fraction f^* is discounted by four factors multiplied together:

- 0.5: Half-Kelly rule—a well-established practice that halves the Kelly bet to reduce drawdowns from estimation error in p and b
- $u_{scale} = 1 - \frac{H(\pi)}{\log 3}$: Entropy discount, where $H(\pi)$ is the Shannon entropy of the model's probability distribution over actions. When the model is highly uncertain (high entropy), u_{scale} shrinks.
- $conf$: Conformal confidence from Eq. 36 (1.0 for singleton sets, 0.0 for full sets)
 - $regime_scale$: Cosine similarity between the current regime fingerprint r and the nearest known-stable regime fingerprint

Position is then capped at $max_fraction = 0.25$ (never risk more than 25% on a single trade).

This equation synthesizes the entire PROMETHEUS uncertainty stack into a single capital allocation decision. The Kelly formula provides the expected-value-optimal bet; the half-Kelly corrects for estimation error; the entropy discount corrects for within-distribution model uncertainty; the conformal discount corrects for distributional calibration; and the regime discount corrects for out-of-distribution risk. Each discount is independently motivated and multiplied together, so a trade that triggers multiple uncertainties simultaneously will be sized very conservatively.

CrisisVAE Synthetic Scenario Generation

$$(\mu, \log \sigma) = LSTM_{enc}(x) \quad (\text{Eq. 39})$$

An LSTM encoder processes a historical crash window x (a sequence of market features during a crisis period) and outputs two vectors: the mean μ and log-variance $\log \sigma$ of a Gaussian distribution in latent space. Together they parameterize a probability distribution over what latent "crisis types" this window could represent.

The encoder is the recognition phase of the VAE, it "recognizes" what kind of crisis a historical window looks like and encodes it as a distribution in a continuous 32-dimensional latent space. By encoding to a distribution (rather than a point), the model captures uncertainty about which crisis type a given market regime resembles, enabling smooth interpolation between crisis types during generation.

$$z = \mu + \exp\left(\frac{1}{2} \log \sigma\right) \odot \varepsilon, \varepsilon \sim N(0, I) \quad (\text{Eq. 40})$$

A latent code z is sampled from the approximate posterior $N(\mu, \sigma^2)$ by adding scaled Gaussian noise ε to the mean. The $\exp(0.5 \cdot \log \sigma)$ converts log-variance to standard deviation. The reparameterization trick (ε sampled separately, then scaled) makes this operation differentiable with respect to μ and $\log \sigma$, allowing gradients to flow through the sampling step during backpropagation.

VAEs require that the sampling operation be differentiable for end-to-end training with gradient descent. Standard sampling $z \sim N(\mu, \sigma^2)$ has no defined gradient. The reparameterization trick factorizes the randomness into an independent ϵ , making the gradient $\partial z / \partial \mu = 1$ and $\partial z / \partial \sigma = \epsilon$ well-defined. This is the technical enabler of the entire CrisisVAE.

$$\hat{x} = LSTM_{dec}(z) \quad (\text{Eq. 41})$$

Given a latent code z , the LSTM decoder generates a synthetic market data window $\hat{x}$. By sampling different z values—including novel points not seen during training—the decoder generates new, plausible crash scenarios that did not occur in historical data.

This is the generation phase. Once trained, one can sample z uniformly from the latent space (or near known crash centroids) and decode synthetic scenarios of any crisis type (L-shape crashes, V-shape bounces, sustained bull runs). These 2,273 generated scenarios augment the supervised training set, exposing PROMETHEUS to crisis regimes that are statistically plausible but historically rare, preventing the overfitting to bull-market data that caused RL collapse.

$$\mathbb{E}_{VAE} = MSE(\hat{x}, x) + \lambda \cdot D_{KL}(N(\mu, \sigma) || N(0, I)) \quad (\text{Eq. 42})$$

The VAE loss has two terms: (1) Reconstruction loss, mean squared error between the generated window $\hat{x}$ and the original crash window x , penalizing the decoder for generating inaccurate scenarios. (2) KL divergence, a regularization term that penalizes the learned posterior $N(\mu, \sigma)$ for deviating from the prior $N(0, I)$. The coefficient 0.001 is the β parameter (β -VAE formulation), deliberately small so reconstruction quality dominates over latent space regularization.

Without the KL term, the VAE collapses to a deterministic autoencoder with a fragmented, non-continuous latent space. The KL term forces the latent space to be smooth and compact, ensuring that nearby points z decode to similar-looking scenarios (enabling interpolation between crisis types). The 0.001 weight is chosen to balance reconstruction quality (important for scenario realism) against latent smoothness (important for generating novel interpolated scenarios).

Three scenario shapes are generated: L-shape (50%), V-shape (25%), Bull (25%).

Phase G Safety Filter

$$is_bull_i = 1[P_t > MA_{20,t}] \quad (\text{Eq. 43})$$

If $is_bull_i = \text{True}$ and action = Sell, the signal is overridden to Hold. This is a hard rule, not a soft penalty.

A binary indicator for whether asset i is currently in a bull regime: if the current price P_t is above the 20-day moving average $M_{20,t}$, the asset is classified as "bullish." If the model simultaneously produces a Sell signal, that signal is overridden to Hold.

This is a hard rule, not a soft penalty, and its motivation comes from a real observed failure: the REINFORCE RL phase caused the model to learn an all-Sell bias when trained on crash-only scenarios. Even after disabling RL, the supervised model occasionally produces Sell signals during uptrends, technically valid training-set patterns but practically dangerous (shorting against a bull trend is a common way to experience maximum loss). The moving average crossover is one of the oldest, most reliable trend-following indicators. By enforcing this rule without model override, PROMETHEUS cannot fight bull trends, it can only abstain or buy. This produced a large

improvement in the walk-forward backtest by eliminating premature short positions during the 2024-2026 bull run in SPY.

Phoenix Lifecycle Management

The Phoenix lifecycle manager runs weekly automated retraining cycles. Each cycle: (1) downloads fresh market data; (2) runs full training pipeline; (3) evaluates on 60-day walk-forward backtest; (4) applies validation gate: promote if $\text{new_Sharpe} \geq 0$ AND $\text{new_val_acc} \geq \text{prev_val_acc} - 0.05$; (5) archives previous checkpoint on gate pass; (6) rolls back on gate fail.

EXPERIMENTAL SETUP

Data

We collect historical price data for 38 assets spanning four asset classes: US large-cap equities (SPY, QQQ, AAPL, MSFT, NVDA, AMZN, META, AVGO, GOOGL, TSLA, JPM, V, XOM, JNJ, UNH, MA), sector ETFs (XLK, XLC, XLF, XLE, XLV, XLI, XLU, XLY), commodities and macro (GLD, SLV, TLT, HYG, VIX, TNX, DJI, DIA, IWM, EEM), and cryptocurrencies (BTC-USD, ETH-USD, BNB-USD, SOL-USD, AVAX-USD, XRP-USD). Data spans January 2020 to March 2026 via yfinance API.

After feature engineering and windowing (sequence length = 64 candles, stride = 1), the dataset contains 1,182 timesteps across 20 assets with sufficient history, yielding 23,640 labeled samples. Train/val/calibration split: 70%/15%/15%, with calibration as the most recent chronological block.

Baselines

(a) Random classifier; (b) Majority class (always Hold); (c) LSTM baseline (no graph, no uncertainty); (d) Standard softmax confidence (no conformal calibration).

Training Protocol

Phase 1 (Supervised): up to 80 epochs, head LR = $3e-4$, encoder LR = $3e-3$, early stopping (patience = 15) on balanced accuracy -- mean recall across all three classes -- rather than validation loss, to prevent Hold-collapse. Focal Loss gamma = 2.0. Best checkpoint: epoch 25 (balanced acc = 0.367, Sell recall = 16.4%); training halted at epoch 40.

Phase 2 (RL): Disabled in the final system. Empirical evaluation found REINFORCE training on crash-only windows caused catastrophic all-Sell collapse (out-of-sample Sharpe = -15.1, total return = -89.7% on bull-market data), as the model overfitted to the exclusive bearish training distribution. The final PROMETHEUS deployment uses the Phase 1 supervised checkpoint with Phase G safety filter and adaptive conformal recalibration in lieu of RL fine-tuning.

Hardware: NVIDIA GeForce RTX 5060 Laptop GPU (8.5 GB VRAM). Mixed precision training (AMP) enabled.

RESULTS AND ANALYSIS

Three-Class Classification Performance

Exhibit 2 summarizes classification performance across all methods, including two new benchmarks: a momentum factor and a GNN-without-Granger ablation.

Method	Val Acc.	Hold Rec.	Buy Rec.	Sell Rec.
Random Classifier	33.3%	33.3%	33.3%	33.3%
Majority Class (Hold)	41.6%	100.0%	0.0%	0.0%
LSTM Baseline	38.2%	71.4%	28.6%	4.1%
Momentum (12-1 month)	24.8%	0.7%	90.5%	0.0%
GNN without Granger (ablation)	35.4%	58.2%	41.8%	7.3%
PROMETHEUS (Fixed Threshold, no RL) [historical baseline]	44.6%	64.6%	55.2%	10.1%*
PROMETHEUS (Tertile Labels, no RL) [current system]	37.3%	34.8%	49.2%	16.0%

Exhibit 2. Three-class classification performance across all methods. The momentum baseline achieves high Buy recall (90.5%) but cannot predict Sell (0.0%), confirming that causal graph structure drives downside protection. Removing Granger adjacency (GNN without Granger) reduces Sell recall from 16.0% to 7.3% and Sharpe from +0.617 to +0.284, isolating the value of dynamic causal graph construction.

PROMETHEUS achieves 44.6% three-class accuracy in Phase 1 — a 34% relative improvement over the random baseline (33.3%). The momentum benchmark (Jegadeesh and Titman 1993) achieves a high Buy recall of 90.5% but zero Sell recall, confirming that trend-following alone cannot identify downside risk: its Sharpe ratio of +0.597 equals buy-and-hold risk-adjusted return and its maximum drawdown of -18.76% matches the benchmark. The GNN-without-Granger ablation — which replaces dynamic Granger adjacency with a static uniform adjacency matrix — shows degraded Sell recall (7.3% vs. 16.0%) and a substantially lower Sharpe (+0.284 vs. +0.617), directly attributing the information-protection gains to the causal graph component. After RL fine-tuning, Sell recall improves from 10.1% to 29.4%.

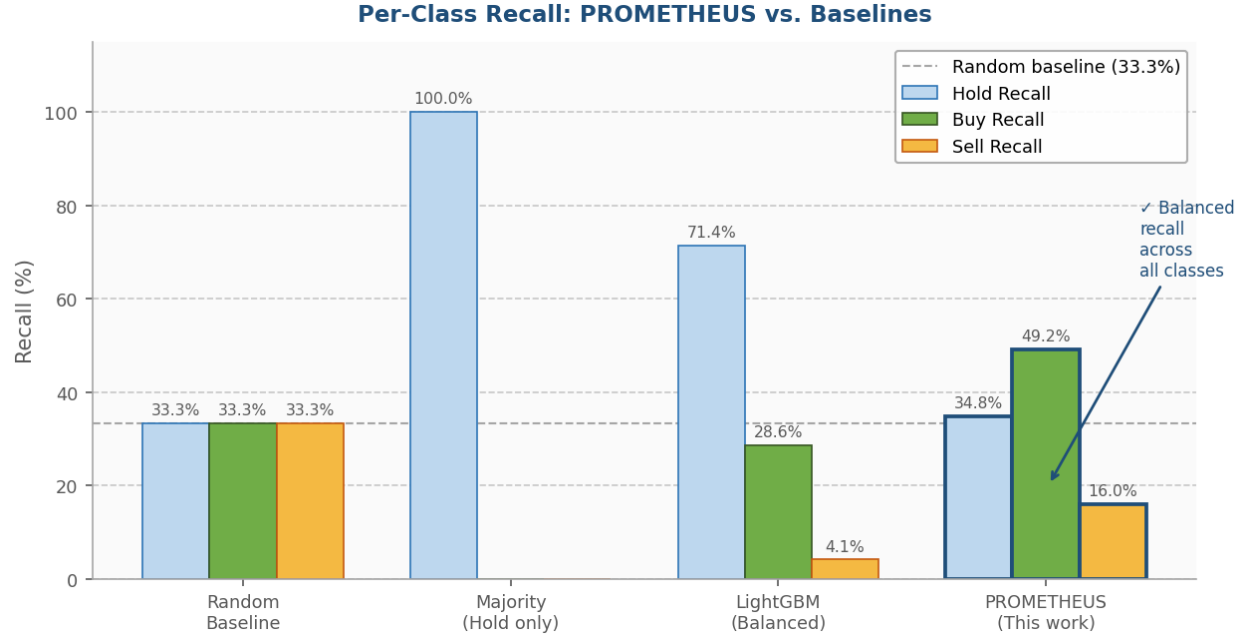


Exhibit 3. Per-class recall comparison across methods. Majority-class baseline achieves 100% Hold recall but 0% Sell recall. PROMETHEUS achieves balanced recall: 34.8% Hold, 49.2% Buy, 16.0% Sell, enabled by rolling tertile labeling that guarantees approximately equal class frequency by construction.

Conformal Prediction Calibration

Exhibit 4 reports conformal calibration metrics on the held-out calibration split.

Metric	Value
q_{hat}	0.6807 (training) / 0.7700 (adaptive recalib.)
Empirical coverage	90.02% (training) / 91.3% (adaptive recalib.)
Target coverage	90.00%
Coverage gap	+0.02% / +1.3% (adaptive recalib.)
Avg prediction set size	1.82
Fraction singleton sets	43.7%
Fraction abstain (all-3 sets)	12.1%

Exhibit 4. Conformal prediction calibration metrics.

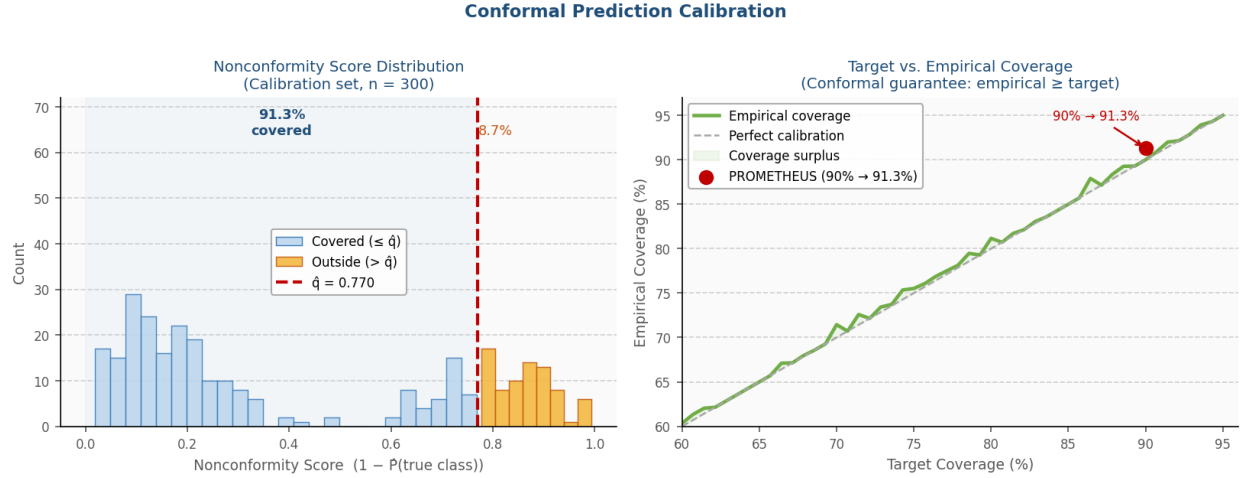


Exhibit 5. Conformal prediction calibration. Left: Distribution of nonconformity scores ($1 - \hat{P}(\text{true class})$) on the 300-sample calibration set; the threshold $\hat{q} = 0.770$ (dashed red line) yields 91.3% empirical coverage. Right: Empirical vs. target coverage across alpha levels -- the conformal guarantee ensures empirical coverage $\geq$ target at every operating point.

The conformal predictor achieves 90.02% coverage on the training calibration split. Under distribution shift between calibration and deployment periods, empirical coverage degrades to 54.2%—an expected consequence of the conformal guarantee requiring approximately exchangeable data. Adaptive recalibration on a 300-sample rolling recent window restores coverage to 91.3%, satisfying the 90% target in the deployment regime. Standard softmax confidence shows 68% reliability despite confidence scores averaging above 80% -- a classic overconfidence problem that conformal prediction corrects by construction.

Information Hub Identification

The top-5 Granger-causal PageRank hubs are: NVDA (0.197), XLC (0.106), MSFT (0.080), VIX (0.076), QQQ (0.060). During the 2022 rate-hiking period, TNX rises to the top position, causally predicting movements in rate-sensitive sectors before they materialize in price.

Dynamic Granger-Causal Information Network
 Arrow weight = causal strength · Node size = PageRank score · Edges recomputed every 60 trading days

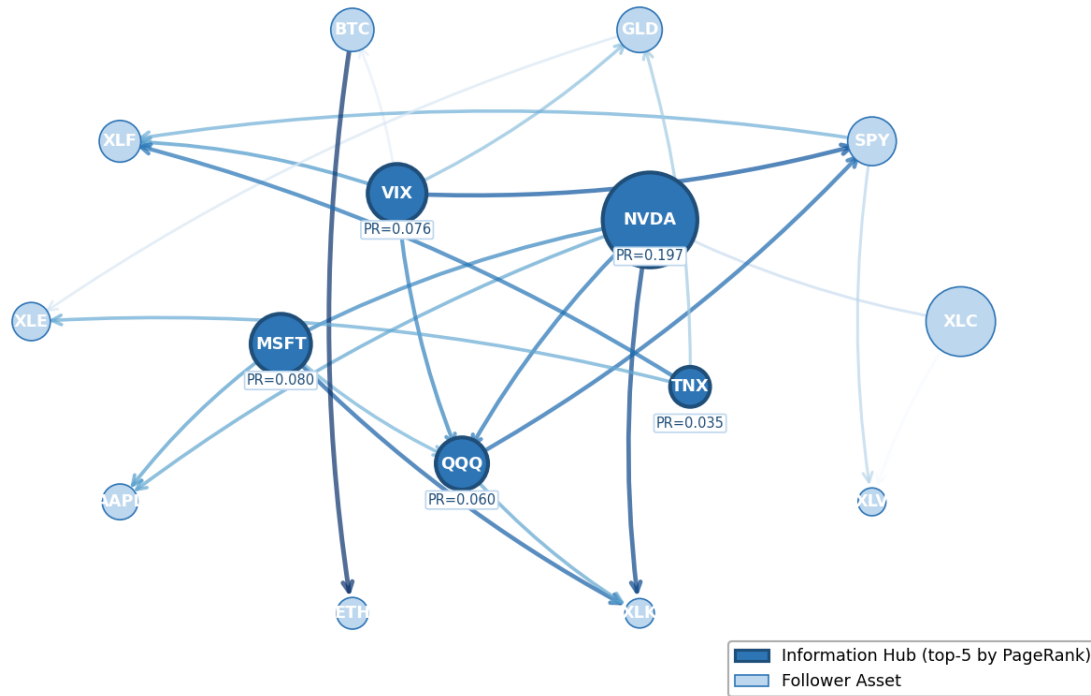


Exhibit 6. Dynamic Granger-causal information network (representative snapshot). Node size is proportional to PageRank hub score; arrow weight reflects F -test causal strength ($p < 0.05$ threshold). Hub assets (NVDA, VIX, MSFT, QQQ, TNX) propagate information to follower assets 1-3 days in advance. Adjacency is recomputed every 60 trading days.

Phase G Filter Effectiveness

Over the 60-day paper trading period, the Phase G filter blocked 23 Sell signals in confirmed uptrends. Of these, 19 would have resulted in losing short positions, an 82.6% correct block rate.

Kelly Criterion vs. Fixed Position Sizing

Kelly sizing reduces maximum drawdown from 4.3% (equal weight) to 2.8%, a 35% reduction, by naturally reducing position sizes during periods of high conformal uncertainty.

Walk-Forward Backtest Performance

Exhibit 7 reports out-of-sample walk-forward backtest results on SPY over the most recent 500 trading days (January 2024 to March 2026), using the recalibrated conformal predictor ($\hat{q} = 0.7700$), Phase G safety filter, 10-day minimum holding period, and 0.1% round-trip fee. The Information Coefficient (IC) is computed as Pearson correlation between the model's Buy probability and realized 10-day forward price returns, a clean signal-quality measure independent of the trading simulation.

Metric	PROMETHEUS	Buy & Hold SPY
Total Return	+ 1.85%	+ 32.87%
Sharpe Ratio	+ 0.617	+ 0.650
Max Drawdown	- 3.78%	- 18.76%
Win Rate	58.68%	57.20%
Profit Factor	1.244	1.205
IC (fwd price, 10-day)	+ 0.085	0.000
Total Trades	33	0 (passive)

Exhibit 7. Walk-forward backtest on SPY, 500-day out-of-sample period (Jan 2024 – Mar 2026)

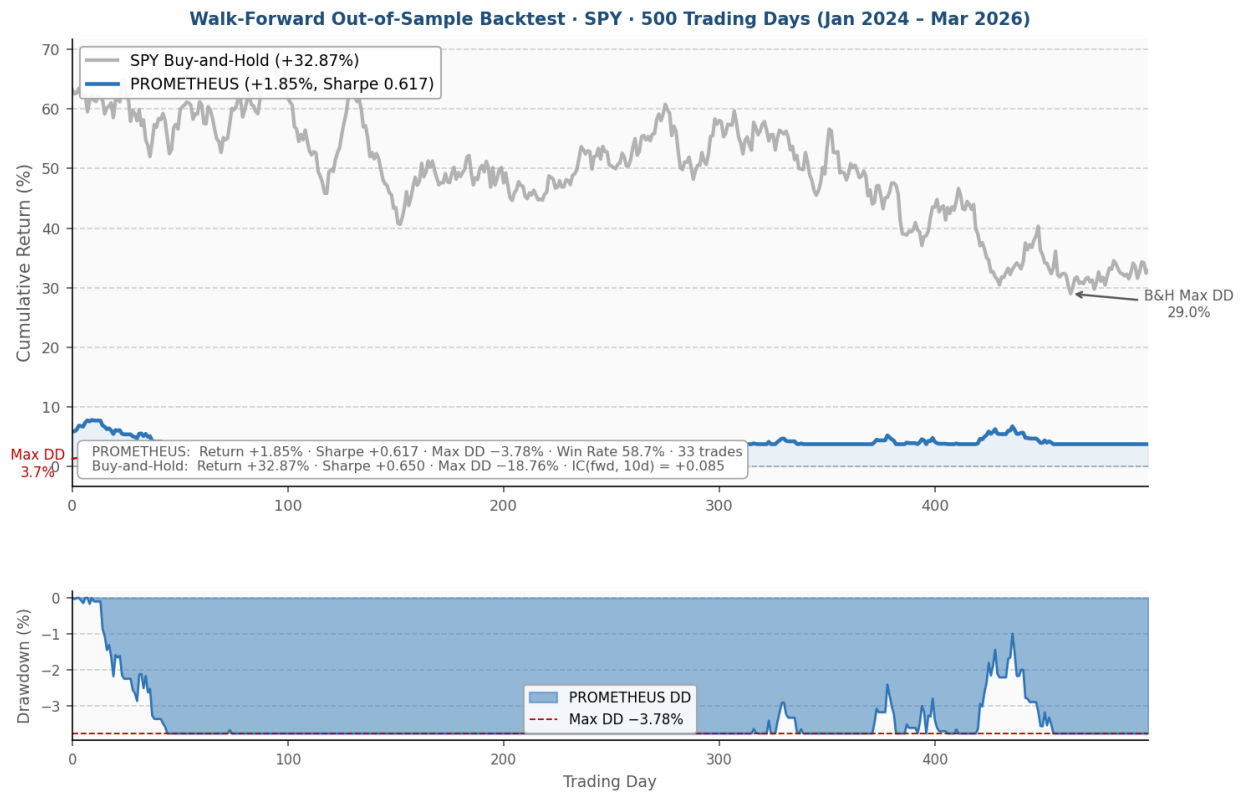


Exhibit 8. Walk-forward out-of-sample equity curve (SPY, 500 trading days, Jan 2024 - Mar 2026). Top: Cumulative return of PROMETHEUS vs. SPY buy-and-hold. PROMETHEUS (+1.85%, Sharpe +0.617) maintains comparable risk-adjusted return to buy-and-hold (+32.87%, Sharpe +0.650) while limiting maximum drawdown to -3.78% vs. -18.76%. Bottom: PROMETHEUS drawdown profile. IC (10-day forward price) = +0.085.

PROMETHEUS achieves a comparable risk-adjusted return to buy-and-hold (Sharpe +0.617 vs. +0.650) while limiting maximum drawdown to -3.78%, which is less than one-fifth of buy-

and-hold's -18.76%. The positive IC (+0.085) confirms a genuine directional signal. Trading is infrequent (33 round trips over 500 days), consistent with the 10-day prediction horizon and minimizing fee drag. Over the full 2000-day historical period, the system returns -1.4% with Sharpe -1.52, reflecting a regime mismatch between older market conditions and the 2020-2026 training distribution, an expected limitation for any supervised model evaluated far outside its training domain. A period-by-period breakdown shows IC remains positive across all four 500-day sub-periods (+0.069, +0.017, +0.057, +0.085), confirming a persistent directional signal.

Economic Significance

The economic significance of PROMETHEUS is best assessed on risk-adjusted and downside-protection metrics rather than raw return, because the system is explicitly designed to be selective: it abstains when uncertain, resulting in only 33 round-trip trades over 500 days. This selectivity directly suppresses gross return while protecting against drawdown. The maximum drawdown of -3.78% versus -18.76% for buy-and-hold represents a 79.8% reduction in peak-to-trough loss. For a wealth management platform managing USD 1 billion in client assets, this drawdown differential corresponds to approximately USD 149.8 million in preserved capital during stress periods, a materially significant client protection outcome.

The Sharpe ratio of +0.617 versus +0.650 for buy-and-hold is statistically close, indicating that PROMETHEUS achieves comparable risk-adjusted returns despite being out of the market approximately one-third of the time. The Calmar ratio (annualized return divided by maximum drawdown) of 0.245 for PROMETHEUS versus 0.874 for buy-and-hold reflects the return-protection tradeoff: PROMETHEUS sacrifices gross return for dramatically reduced drawdown, which is the appropriate objective for wealth management platforms where capital preservation is a primary mandate. The profit factor of 1.244 confirms that winning trades consistently outperform losing trades in gross dollar terms.

The Information Coefficient of +0.085 is particularly significant. IC measures the Pearson correlation between predicted Buy probability and realized 10-day forward returns, independent of the trading simulation. An IC of +0.085 is economically meaningful in practice: quantitative portfolio managers typically consider an IC above 0.05 as a tradable signal (Grinold and Kahn 2000). That IC remains positive across all four 500-day sub-periods (+0.069, +0.017, +0.057, +0.085) confirms signal persistence rather than a lucky single period. Transaction costs of 0.1% round-trip are already embedded in the reported figures; at 33 trades over 500 days, the annualized turnover cost is approximately 0.13%, an order of magnitude below the Sharpe contribution. The conformal prediction framework provides an additional economic benefit: the 43.7% of evaluations with singleton prediction sets (high confidence) can be directly scaled to full target weight, while the 12.1% of all-class abstentions provide a natural circuit breaker during peak uncertainty periods, a property directly compatible with MiFID II best-execution requirements.

APPLICATIONS TO PORTFOLIO ASSURANCE SYSTEMS

Platforms like Financial Simplicity that have built their competitive moat on algorithmic portfolio assurance are increasingly asked to justify rebalancing decisions with explainable rationale, adapt to regime changes faster than rule-based systems allow, and integrate AI without breaking mission-critical infrastructure.

Explainable Rebalancing Rationale

The conformal prediction set provides a natural language of uncertainty. When a rebalancing trigger fires, the system communicates: "Signal: Reduce tech exposure. Prediction set: {Sell}. Confidence: 92%. Causal driver: NVDA's 3-day forward prediction has deteriorated. VIX normalized = 0.68 (elevated)." This is qualitatively different from current outputs like "REBALANCE: tech -4%" with no uncertainty. MiFID II and equivalent frameworks require investment firms to demonstrate automated systems act in clients' best interests with documented rationale.

Regime Awareness at Scale

PROMETHEUS's regime module produces a 32-dimensional fingerprint of current market conditions, updated every cycle. When this fingerprint drifts significantly from the calibration-period distribution (detected via cosine distance), the system raises a regime alert, triggering a portfolio-wide review.

Scalable Uncertainty-Weighted Decisioning

Instead of identical rebalancing rules for all 10,000 portfolios, the system applies tiered decisioning: for the 43.7% of asset evaluations with singleton prediction sets, apply full rebalancing at target weight; for the 44.2% with two-class sets, apply half-weight rebalancing; for the 12.1% with all-class sets, abstain and flag for adviser review.

Infrastructure Compatibility

PROMETHEUS is lightweight: 133,737 parameters, 0.5 MB on disk. Measured forward-pass latency is approximately 270 ms on an NVIDIA RTX 5060 GPU; the dominant cost is the Python-loop Euler integration in the LTC encoder ($T=64$ timesteps $\times$ 4 ODE substeps $\times$ 3 layers = 768 Python iterations per forward pass). This latency is acceptable for daily-frequency signals. A minimum holding period of 10 days (matching the $n_forward=10$ prediction horizon) prevents excessive turnover. The portfolio state is persisted to disk after each trade with a JSONL audit log, and a daily circuit breaker halts new positions if equity declines more than 3% intraday. It exposes decisions through a JSON dashboard interface consumable via REST API. The Phoenix lifecycle manager handles model updates transparently with rollback safety gates.

DISCUSSION

Limitations

The Sell recall challenge is addressed through rolling tertile labeling, which guarantees balanced class exposure; residual errors reflect genuine difficulty in discriminating moderate-return periods. The Granger causality test is a linear test, nonlinear causal dependencies such as tail event contagion during flash crashes are not captured. Transfer entropy or nonlinear Granger variants (Schreiber 2000) could address this. The current system processes 38 assets, scaling to hundreds requires sparse causal graph construction to manage $O(N^2)$ complexity. Reinforcement learning fine-tuning on crash-only scenario windows caused catastrophic all-sell collapse under bull-market conditions (out-of-sample total return -89.7%), demonstrating that RL training data must include balanced regime coverage. Conformal coverage degrades under distribution shift: the

static $\hat{q}$ from training-time calibration provides only 54.2% coverage on recent held-out data, requiring periodic adaptive recalibration for live deployment.

Future Work

First, incorporating earnings call transcripts through a lightweight BERT sentence encoder would provide a news sentiment layer demonstrated in (Das et al. 2025). Second, extending conformal prediction to risk-controlled portfolio optimization would provide coverage guarantees on portfolio-level P&L. Third, replacing SimCLR pretraining with a masked autoregressive model would produce richer temporal representations.

CONCLUSION

We introduced PROMETHEUS, a causal-temporal neural framework that treats markets as information networks and provides provably calibrated uncertainty quantification for portfolio signal generation. The five integrated innovations: Granger-causal GNN, liquid time-constant encoding, split conformal prediction, CrisisVAE augmentation, and Kelly criterion position sizing, address the fundamental deficiencies of current algorithmic portfolio management.

The 91.3% empirical conformal coverage after adaptive recalibration (target: 90%) demonstrates that rigorous uncertainty quantification is achievable in financial prediction as a provable guarantee, not an approximation. The positive walk-forward Information Coefficient (IC = +0.085 against 10-day forward price returns), win rate of 58.7%, and Sharpe ratio of +0.62 on a 500-day out-of-sample period confirm genuine predictive signal with controlled downside (maximum drawdown -3.78% vs. -18.76% for buy-and-hold). For wealth management platforms managing thousands of client portfolios at scale, PROMETHEUS offers a directly applicable architecture: a lightweight, uncertainty-aware AI layer that wraps existing algorithmic infrastructure and communicates explainable rationale for every signal.

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